VII.

Reference solution

The following equations

give the electron's total energy:

The condition gives hydrogen's orbital radius and speed:

The Larmor formula gives the initial radiated power:

During a short interval , radiation reduces the electron's total energy by

Here is the decrease in orbital radius. The relation between time and radius is

where

For a sphere of radius , is the volume of a thin shell at radius . Summing from to gives volume . Correspondingly, the electron orbit shrinks from to in a time